API Protocol: MoodsCfg

Duotecno	Showroom:	T +3250 200560
Astridlaan 170A	Ten Briele 18.1	F +3250 200561
8310 Assebroek	8200 Sint-Michiels	
Belgium	Belgium	email: info@duotecno.be

Domotica | Audio | Video

API PROTOCOL MOODSCFG

Version 1.15

Info: http://www.duotecno.be

Overview:

This API can be used for the mood configuration in a Duotecno node.

The base url for this API is: **[IP-Address]:8080/**

[IP-Address]: The IP-Address of the Duotecno node.
The port is always 8080.

Possibilities:
- Request tag, time and date of last changes.
- Full mood configuration: Upload / Download / Erase / Request summary.
- Single mood configuration: Request / Edit / Erase / Delete

NOTE:
When the tag has changed FIRST download the current configuration of the mood(s). Edit the needed values and upload the mood(s) again.

Json data objects - Overview

Json Data object - mood

This object can be separated in a header and a data part. The header part holds info about the mood. The data part holds the functions to be executed when the mood is invoked.

When requesting mood info there is always a mood header available for every virtual unit.

Key	Value	Description
assigned	Boolean	Mood header – Read only This value indicates if the mood is assigned to a virtual unit. This key has no meaning when editing or uploading mood data.
data	Array	Mood data – Read/Write Array of data entry objects. These are the finctions that are executed when the mood is activated. When there are no entries this value is null or an empty json array.
description	String	Mood header – Read/Write Optional: A text describing the function of the mood. This text should be displayed to the end-user and gives information about the scene. When there is no text description this field is omitted.
empty	Boolean	Mood header – Read only This value indicates if there are data items available. This property is meaningful when a summary of the moods is requested. This key has no meaning when editing or uploading mood data.
id	integer	Mood header – Read/Write The identifier for the mood. This is also the unit address of the virtual unit.
name	String	Mood header – Read/Write The name of the vitual unit (=Mood).
userAdjustable	Boolean	Mood header – Read/Write

		This value indicates if the end-user can change the mood. Moods that the end-user cannot change should not be displayed in the end-user configuration.

Json Data object – mood entry data

This object holds the functions to be executed when the mood is invoked.

Key	Value	Description
delay	int	Optional: Only when there is a delay. The value is a multiplier for 60 msec. The value should be limited from 4 – 83. (250 msec – 5000 msec)
msg	String	The data message. See Appendix: Data messages.
nodeAddress	Integer	The node address for the destination unit.
unitAddress	Integer	The unit address for the destination unit.

Json Data object – moods

The moods object is used for full configuration.

Key	Value	Description
moods	Array	Array of Mood objects.

Json Data object – moods_info

The moods_info object is used when a summary of the configuration is requested.

Key	Value	Description
Moods_info	Array	Array of Mood header objects.

Json data object – database tag info

This info holds a tag, date and time when the last changes were made.

Key	Value	Description
date	string	Optional: The date of last changes. “dd.mm.yyyy”
time	String	Optional: The time of last changes. “hh:mm:ss”.
tag	String	A 20 char tag. The first time requested or when there is no configuration (after a factory init) this can be an empty string.

Json Data object – Status message

Json responses are returned with http status code 200. They are a response to a POST/PUT http message that holds a Json data object in the http body.

Json responses always hold a “status” key.
When the value is “ok” then the http request was a success.

```
{
	"status": "ok"
}
```

When the value is “error” then an error happened and optional the “error” and “error_detail” key are added:

```
{
	"error": "invalid_request",
	"error_detail": "Invalid refresh token",
	"status": "error"
}
```

Request tag, time and date last changes

When the configuration is changed the time, date and a tag are generated and stored.
These values can be requested by the API.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.29

Overview:

http method	GET
Endpoint	/moodscfg/tag
Response	Status code 200 + Json database taginfo object

Response:


```
{
        “date": "07.10.2020",
        "tag": "YRjkNSBSDlrVcPtvtvHr",
        "time": "16:22:00"
}
```


The response can be empty tag value.


```
{
        "tag" : ""
}
```


This is possible when:

- There is no mood configuration present.
- Configuration has not been changed since a firmware update supporting this API.

Download the full configuration

Download the full mood configuration from the node.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.29

Overview:

http method	GET
Endpoint	/config/download/moods.json
Response	Status code 200 + Json moods data object

Response:

HTTP Status code 200 + Json moods data object:


```
{
    "moods": [
     {
        "assigned": true,
         "data": [
          {
              "delay": 4,
              "msg": "A2020401",
              "nodeAddress": 36,
              "unitAddress": 0
           }, … ]
        “description” : “This mood is controlled by the upper left button on the switch in the kitchen”,
        "empty": true,
        "id": 20,
        "name": "Mood 0",
        "userAdjustable": true
     },  …
     ]
}
```


Key	Value	Description
assigned	Boolean	Mood header – Read only The value is true when this mood is created and is assigned to a virtual unit. When this value is false then there is no mood (info) assigned to this virtual unit but a mood can be created and attached to this unit. Note: A node can hold different types of units. Moods can only be assigned to a virtual unit. When the unit is a different type it is omitted in this response.
description	String	Mood header – Read/Write Optional: A text describing the function of the mood. When there is no text this field is omitted.
empty	Boolean	Mood header – Read only The value is true when there are no data items present. This is an empty mood with only a header.
id	integer	Mood header – Read/Write The identifier for the mood. This is the unit address of the virtual unit.
name	String	Mood header – Read/Write The name of the vitual unit (=Mood).
userAdjustable	Boolean	Mood header – Read/Write Flag to indicate if the end-user can change the mood.

Key	Value	Description
delay	int	The value is a multiplier for 60 msec. The value should be limited from 4 – 83. (250 msec – 5000 msec) When this value is omitted the default value is 0.

msg	String	The data message. See Appendix: Data messages. This value cannot be empty or null.
nodeAddress	Integer	The node address for the destination unit. This value cannot be empty or null.
unitAddress	Integer	The unit address for the destination unit. This value cannot be empty or null.

Upload the full configuration:

Upload the full mood configuration to the node.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.29

Overview:

http method	POST
Endpoint	/config/upload/moods.json
Data body	Json moods data object
Response	Status code 200 + Json status object

Request Json data body:

```
{
    "moods": [
     {
        "data": [
         {
            "delay": 4,
            "msg": "A2020401",
            "nodeAddress": 36,
            "unitAddress": 0
          }, … ]
       “description” : “This mood is controlled by the upper left button on the switch in the kitchen”,
       "id": 20,
       "name": "Mood 0",
       "userAdjustable": true
     },  …
     ]
}
```

Key	Value	Description
data	Array	Mood data Array of mood data entry objects. To erase the entries set the value to null or an empty json array.

		NOTE: When set to null an empty mood with only a header is created and assigned to a virtual unit.
description	String	Mood header A text describing the function of the mood. This text should be displayed to the end-user. When this key is omitted the description is erased. When set to “null” or empty string it is also erased.
id	integer	Mood header – Read/Write The identifier of the mood. This is the unit address of the virtual unit.
name	String	Mood header – Read/Write The name of the virtual unit (=Mood). When the value is omitted or set to null the value is ignored and the name is not changed. To clear the name use. “name”:””
userAdjustable	Boolean	Mood header. Flag to indicate if the end-user can change the mood. When this key is omitted this value is defaulted to false.

Key	Value	Description
delay	int	The value is a multiplier for 60 msec. The value should be limited from 4 – 83. (250 msec – 5000 msec) When this value is omitted the default value is 0.
msg	String	The data message. See Appendix: Data messages. This value cannot be empty or null.
nodeAddress	Integer	The node address for the destination unit. This value cannot be empty or null.
unitAddress	Integer	The unit address for the destination unit. This value cannot be empty or null.

Erase the full configuration:

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.29

Overview:

http method	POST
Endpoint	/config/upload/moods.json
Data body	Json moods data object
Response	Status code 200 + Json status object

Request Json data body:

```
{
        “moods” = [ ]
}
```

Request Mood configuration summary.

This request can be used to get a summary of all the available moods.

The end-user configuration app can use this request to get info from all the moods. The moods which are not user adjustable should not be displayed.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.29

Overview:

http method	GET
Endpoint	/moodscfg/info
Response	Status code 200 + Json moods_info object

Response:

```
{
  "moods_info": [
    {
      "assigned": true,
      “description” : “This mood can be controlled on the touchscreen – needs to edited”,
      "empty": true,
      "id": 20,
      "name": "Mood 0",
      "userAdjustable": true
    },
    {
      "assigned": true,
      "empty": true,
      "id": 21,
      "name": "Mood 2",
      "userAdjustable": false
    }, …
  ]
}
```

Key	Value	Description
assigned	Boolean	Mood header – Read only The value is true when this mood is created and is assigned to a virtual unit. When this value is false then there is no mood (info) assigned to this virtual unit but a mood can be created and attached to this unit. Note: A node can hold different types of units. Moods can only be assigned to a virtual unit. When the unit is a different type it is omitted in this response.

description	String	Mood header – Read/Write Optional: A text describing the function of the mood. When there is no text this field is omitted.
empty	Boolean	Mood header – Read only The value is true when there are no data items present. This is an empty mood with only a header.
id	integer	Mood header – Read/Write The identifier for the mood. This is the unit address of the virtual unit.
name	String	Mood header – Read/Write The name of the vitual unit (=Mood).
userAdjustable	Boolean	Mood header – Read/Write Flag to indicate if the end-user can change the mood.

Single mood: Request

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.29

Overview:

http method	GET
Endpoint	/moodscfg/[ID]
Response	Status code 200 + Json mood object

[ID] : Identifier (unitAddress).

Single mood: Create / edit

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.29

Overview:

http method	POST
Endpoint	/moodscfg/
Data body	Json mood data object
Response	Status code 200 + Json status object

Request Json data body:

```
{
        "data": [
          {
            "delay": 4,
```

```
            "msg": "A2020401",
            "nodeAddress": 36,
            "unitAddress": 0
          }
        ],
        "description" : "Living Spots on",
         "empty": false,
         "name" : "Living spots",
         "id": 5,
         "userAdjustable": true
}
```

The fields are the same as when uploading the full configuration.

Single mood: Erase data entries.

This can be used to erase the data entries of the mood but still keep the header info.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.29

Overview:

http method	POST
Endpoint	/moodscfg/
Data body	Json mood data object
Response	Status code 200 + Json status object

Request Json data body:

```
{
        "data": [ ]
        "description" : "This mood is controlled by the upper left button on the switch in the kitchen",
        "id": 20,
        "name": "Mood 0",
        "userAdjustable": true
}
```

Single mood: Delete

This function deletes the mood header & data.
After invoking this function there Is no mood data assigned to the virtual unit.

Supported nodes:

Node	Minimum firmware version

Touchscreen	V66.02.29

Overview:

http method	DELETE
Endpoint	/moodscfg/[ID]
Response	Status code 200 + Json status object

[ID] : Identifier (unitAddress).

Appendix: Data message:

All data is in hex integer values. Every data-byte is 2 characters.

msgCode	DataLength	Data0 (Method)	Data 1	Data 2	Data 3	Data 4	Data 5

The data length is maximum 6 data bytes long.
The data Length field is the number of data bytes. (excluding the message code.)

Dimmer unit:

Message code: A2

	Method (hex)	Data 1	Data 2	Data 3	Data 4	Data 5
Set On/Off	04	0 or 1				
Set Dim Up	05					
Set Dim Down	06					
Set Dim value	03	Value				
Set PIR ON	13	Time	Value			

Time in minutes: 01 – 3C or FF (1 – 60 minutes , FF = 3minuten)
Value: 01 – 63 or FF (1-99, FF = Current value)

<u>Examples:</u>

Set dimmer Off	"A2020400"
Set dimmer On	"A2020401"
Set dimmer dim up	"A20105"
Set dimmer value 50%	"A2020332"
Set Dimmer PIR on – 3min – 50%	"A203130332"

Switch unit:

Message code: A3

	Method (hex)	Data 1	Data 2	Data 3	Data 4	Data 5
Set On/Off	01	0 or 1				
Set PIR ON	07	Time				

Time in minutes: 01 – 3C or FF (1 – 60 minutes , FF = 3minuten)

<u>Examples:</u>

Set switch Off	"A3020100"
Set switch On	"A3020101"
Set switch Pir On – 5 minutes	"A3020705"

Control unit:

Message code: A8

	Method (hex)	Data 1	Data 2	Data 3	Data 4	Data 5
Set Pulstoggle	03	0 or 1				
Set Toggle (Long)	04	0 or 1				

<u>Examples:</u>

Set Pulstoggle 1 "A8020301"
Set Toggle 0 "A8020400"

LCD Virtual unit:

Message code: A6

	Method (hex)	Data 1	Data 2	Data 3	Data 4	Data 5
Set Pulstoggle	03	0 or 1				
Set Toggle (Long)	04	0 or 1				

<u>Examples:</u>

Set Pulstoggle 1 "A6020301"
Set Toggle 0 "A6020400"

Duoswitch unit:

Message code: B6

	Method (hex)	Data 1	Data 2	Data 3	Data 4	Data 5
Set Up	04					
Set Down	05					
Set Stop	03					

Examples:

Set Up "B60104"
Set Down "B60105"
Set Stop "B60103"

Sensor unit:

Message code: 88

	Method (hex)	Data 1	Data 2	Data 3	Data 4	Data 5
Set on/off	03	0 or 1				
Set Setpoint	01	SP MSB	SP LSB	Preset		
Set Preset	0D	Preset				
Set working mode	10	Work Mode				
Set Fan Speed	11	Speed				
Set Swing Angle	12	Angle MSB	Angle LSB			
Set Swing mode	13	Swing Mode				

Setpoint: Values * 10.

- 21°C → 21*10 = 210 -> 0x00D2
- 27,5°C → 27,5 * 10 = 275 -> 0x0113

Preset: 0-3 (00= sun / 01 = half sun / 02 = moon / 03 = half-moon).
Work mode / Fan speed / Swing angle / Swing mode: these values depend on the system being used. Can be found in the document "TCP/IP open protocol": Appendix 1: Extended sensor unit functions

Examples:

Set setpoint sun 21°C	"88040100D200"
Set setpoint Half sun 27,5°C	"880401011301"
Set preset sun	"88020D00"

Audio extended unit:

Message code: 9F

	Method (hex)	Data 1	Data 2	Data 3	Data 4	Data 5
Set Off	05	Dest				
Set Source	01	Dest	Source			
Set Volume Up	08	Dest				
Set volume dn	09	Dest				
Set volume stop	0A	Dest				
Set volume	0C	Dest	Volume			
Set Func Play	0F	Dest				
Set Func Stop	10	Dest				
Set Func Pause	11	Dest				
Set Func FF	12	Dest				
Set Func RR	13	Dest				

Set Func FFWD	14	Dest				
Set Func FRWD	15	Dest				
Set Extra IR	18	Dest	Extra_ func			

Dest: 00-03
Source: 00-07
Volume: 00 – Max value (Max value depends on A/V system - Check Json Configuration files.)

Extra source functions: (extra_func)

Key	Value	Description
KP_1	0x00	Keypad button 1
KP_2	0x01	Keypad button 2
KP_3	0x02	Keypad button 3
KP_4	0x03	Keypad button 4
KP_5	0x04	Keypad button 5
KP_6	0x05	Keypad button 6
KP_7	0x06	Keypad button 7
KP_8	0x07	Keypad button 8
KP_9	0x08	Keypad button 9
KP_0	0x09	Keypad button 0
KP_UP	0x0A	Keypad menu up
KP_DOWN	0x0B	Keypad menu down
KP_LEFT	0x0C	Keypad menu left
KP_RIGHT	0x0D	Keypad menu right
KP_ENTER	0x0E	Keypad menu enter
KP_10	0x0F	Keypad button -/--
KP_FREE1	0x10	Keypad free function 1
KP_FREE2	0x11	Keypad free function 2
KP_FREE3	0x12	Keypad free function 3
KP_FREE4	0x13	Keypad free function 4
KP_FREE5	0x14	Keypad free function 5
UP1	0x15	Keypad extra up function
DOWN1	0x16	Keypad extra down function
UP2	0x17	Keypad extra up function
DOWN2	0x18	Keypad extra down function

Examples:
Set destination 1 – Source 5 “9F03010105”
Set destination 3 volume up “9F020803”
Set destination 0 Keypad up “9F0318000A”

Audio basic: (deprecated)

Message code: 9F

	Method	Data 1	Data 2	Data 3	Data 4	Data 5

	(hex)					
Set Off	05					
Set Source	01	Source				
Set volume up	08					
Set volume dn	09					
Set src func	0E	0=play 1=stop				

Source: 00-07

Audio Bose unit:

Message code: 9F

	Method (hex)	Data 1	Data 2	Data 3	Data 4	Data 5
Set Off	05					
Set Source	01	Source				
Set volume up	08					
Set volume dn	09					
Set src func	0E	0=play 1=stop				
Extra source func (**)	17	Extra_ Func				
Play (**)	20					
Stop (**)	21					
Pause (**)	22					
Next (**)	23					
Prev (**)	24					
FFWD (**)	25					
FRWD (**)	26					

(**): Only for Bose V35/535 systems.

Source: 00-07
- 00: src0/fm ,
- 01: src1/hdmi1,
- 02: src2/hdmi2,
- 03: src3/hdmi3,
- 04: src4/in1,
- 05: src5/in2,
- 06:src6/ipod,
- 07:src7/local

Extra source functions: (extra_func)

Method	Value	Description
Keypad 1	0x00	Keypad digit 1
Keypad 2	0x01	Keypad digit 2
Keypad 3	0x02	Keypad digit 3

Keypad 4	0x03	Keypad digit 4
Keypad 5	0x04	Keypad digit 5
Keypad 6	0x05	Keypad digit 6
Keypad 7	0x06	Keypad digit 7
Keypad 8	0x07	Keypad digit 8
Keypad 9	0x08	Keypad digit 9
Keypad 0	0x09	Keypad digit 0
Keypad up	0x0A	Keypad up
Keypad down	0x0B	Keypad down
Keypad left	0x0C	Keypad left
Keypad right	0x0D	Keypad right
Keypad enter	0x0E	Keypad enter
Keypad dash	0x0F	Keypad dash
Dvd menu	0x10	DVD menu
Guide	0x11	Guide
Info	0x12	Info
Exit	0x13	Exit menu
free	0x14	Not used - reserved
Up1	0x15	Red (Only Bose V535 and newer systems)
Down1	0x16	Green (Only Bose V535 and newer systems)
Up2	0x17	Yellow (Only Bose V535 and newer systems)
Down2	0x18	Blue (Only Bose V535 and newer systems)

IRTX unit:

Message code: AD

	Method (hex)	Data 1	Data 2	Data 3	Data 4	Data 5
Set puls	03	MSB	LSB			

Examples:
Send IRCode 7 “AD03030007”
Send IRCode 300 “AD0303012C”

AV-Matrix unit:

Message code: CA

	Method (hex)	Data 1	Data 2	Data 3	Data 4	Data 5
Set group master	14	Output	Master	Slave		
Add slave to group	15	Output	Master	Slave		
Remove slave from group	16	Output	Master	Slave		

Output / Master / Slave values depend on type of av-matrix unit.

Examples:

Create group master output0 / slave 1	“CA0414000001”
group master output0 / Add slave 2	“CA0415000002”